

Supplement to Repair What Matters

Swayam Singal

Supplement: Gaussian finite-edit display bound

For one protected pixel, let U be the unchanged ordered Gaussian support and E the complete edited support. Effective per-splat alpha is assumed to be post-clamp/post-cutoff and to satisfy $\alpha_i \in [0, 1]$. Per-channel emitted color after spherical-harmonic evaluation and the background lie in the same interval of width C_{color} .

For edited support E , define

$$A(E) = 1 - \prod_{i \in E} (1 - \alpha_i).$$

This is the maximum fraction of final transmittance whose contribution can be replaced, attenuated, or revealed through the edited splats. Since any channel value differs by at most C_{color} ,

$$\|R(U \cup E) - R(U)\|_{\infty} \leq C_{\text{color}} A(E).$$

For before/after edited sets E_0, E_1 , insert the common unchanged state U and apply the triangle inequality:

$$\begin{aligned} \|R(U \cup E_0) - R(U \cup E_1)\|_{\infty} &\leq \|R(U \cup E_0) - R(U)\|_{\infty} \\ &\quad + \|R(U) - R(U \cup E_1)\|_{\infty} \\ &\leq C_{\text{color}} (A(E_0) + A(E_1)). \end{aligned}$$

The output channel itself cannot change by more than its admissible interval width, hence

$$\|R(U \cup E_0) - R(U \cup E_1)\|_{\infty} \leq C_{\text{color}} \min\{1, A(E_0) + A(E_1)\}.$$

Insertion and deletion are obtained by taking one edited set to be empty. Depth-order changes involving edited splats are included because the edited set must contain every Gaussian whose projected opacity, color, or order relation can change. Renderer behavior outside these bounded assumptions is not softened into the certificate; it becomes an exact dependency or triggers FULL fallback. The image-space certificate is the maximum of the pixel bound over the declared protected support.

Opacity-only delta refinement used by v5–v6

The general finite-edit bound above is deliberately indifferent to edit magnitude once the edited support is known. This is safe for geometry, color, insertion, deletion, and ordering changes, but it can be unnecessarily loose when corresponding Gaussian records differ only in opacity. The confirmatory v5–v6 path therefore uses a narrower bound under stronger invariants.

Consider front-to-back compositing at one protected pixel. Let $\alpha_i^0, \alpha_i^1 \in [0, 1]$ be the effective renderer-admitted alpha values for paired Gaussian i before and after the revision. Position, projected covariance, source-order depth, rotation, and spherical-harmonic color are identical for

each pair. Denote the accumulated color after processing splats $1, \dots, n$ by C_n . Changing one alpha coordinate while holding all other coordinates fixed can replace at most the remaining channel range multiplied by the alpha change. If every possible emitted/background channel lies in an interval of width C_{color} , then

$$\|C_n(\dots, \alpha_i^0, \dots) - C_n(\dots, \alpha_i^1, \dots)\|_\infty \leq C_{\text{color}} |\alpha_i^0 - \alpha_i^1|.$$

Applying this coordinate-wise along a path from the full before alpha vector to the full after alpha vector and using the triangle inequality gives

$$\Delta_p \leq C_{\text{color}} \sum_i |\alpha_i^0(p) - \alpha_i^1(p)|.$$

The renderer used by the independent oracle may stop compositing once accumulated alpha exceeds 0.999. Relative to the corresponding non-terminated compositor, either endpoint can then omit at most 0.001 transmittance. A before/after difference therefore needs at most an additional $0.002C_{\text{color}}$ allowance. The implemented pixel certificate is

$$B_p^{\text{opacity}} = C_{\text{color}} \min\left(1, \sum_i |\alpha_i^0(p) - \alpha_i^1(p)| + 0.002\right)$$

whenever at least one admitted alpha changes at p . If the summed alpha variation is exactly zero, the implementation returns an exact zero bound instead of adding the early-termination allowance globally. This preserves zero support outside the affected pixels.

The refinement is intentionally fail-closed. Paired primitive counts and spherical-harmonic degree must agree, and every paired Gaussian must be identical except for opacity logit. A geometry, covariance, rotation, spherical-harmonic, or unexpected visibility change rejects the refinement and returns control to the general finite-edit certificate or to full fallback. The v5 mechanism experiment and v6 confirmatory breadth experiment use this opacity-only path without changing the general translation control.

Generative AI Use Disclosure

OpenAI ChatGPT was used as an assistive tool for portions of software and scripting development, debugging, experimental planning, and drafting and language editing associated with this work. The reported measurements and evidence were produced by the project software and frozen experimental artifacts rather than generated by the language model. The author retains full responsibility for the code, analysis, claims, conclusions, and final text.